HERIOT-WATT UNIVERSITY

F79MB Statistical Models B

Assessment Project 2 – 2024: Sample solution

1. (a) Summary statistics for Purchase amount are:

```
   Min     1st Qu.  Median    Mean    3rd Qu.    Max.
 74.64     148.30   170.74    177.59  201.56     324.37
```

with a standard deviation of `46.79` dollars.

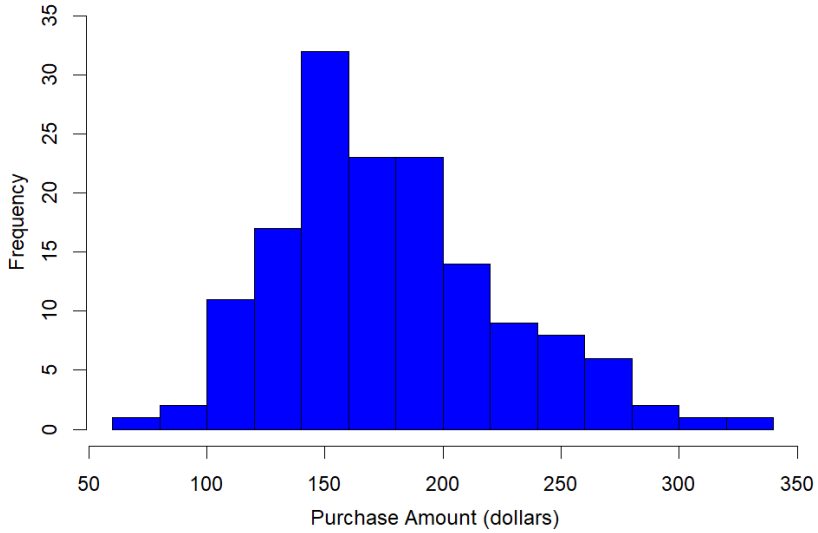


Figure 1: Histogram of the purchase amount of a sample 150 customers.

The histogram of Purchase amount in Figure 1 shows a unimodal, right-skewed distribution. This is consistent with the upper quartile is further than the lower quartile from the median, and that the mean is greater than the median.

Summary statistics for Age are:

```
Min.   1st Qu.  Median  Mean   3rd Qu.  Max.
18.00   33.25   45.00   45.11   59.00   70.00
```

with a standard deviation of `14.85525` years.

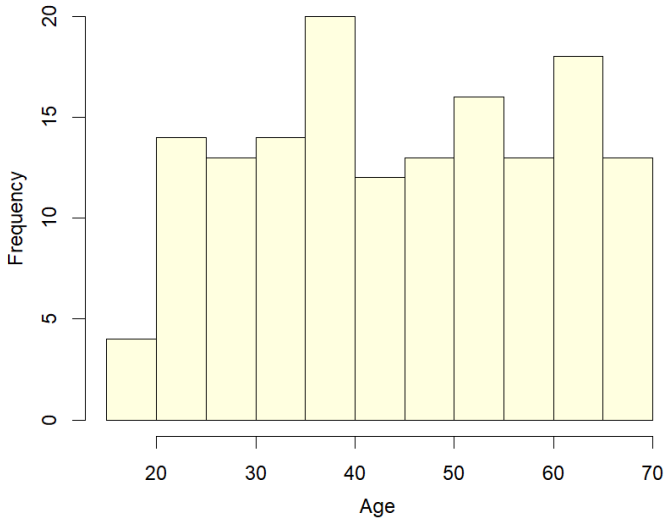


Figure 2: Histogram of age of a sample 150 customers.

The histogram of Age in Figure 2 is roughly uniformly distributed with a minimum age of 18 and maximum age of 70. The standard deviation is 14.86.

Table 1 presents a summary of 'Membership Level': There are a total of 150 membership cards, with 39.3% Gold, 30.7% Platinum, and 30% Silver. The data primarily consists of 'Gold' membership members, with 'Platinum' and 'Silver' members being nearly equal in number (see Figure 3).

Membership Level	Gold	Platinum	Silver
Frequency (%)	59 (39.3)	46 (30.7)	45 (30)

Table 1: Frequency table of membership level.

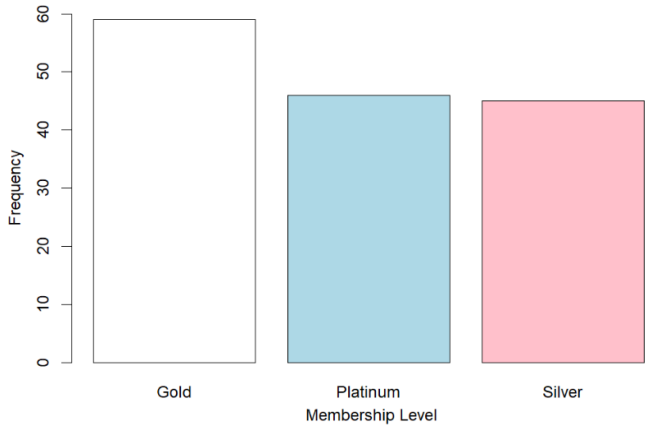


Figure 3: Bar plot of membership level a sample 150 customers.

1(b) In Figure 4, there is an apparent increasing relationship, possibly linear, between purchase amount and age, but with a lot of random scatters. It looks like those with Gold membership tend to purchase more. As shown in the plot, Gold members spend more than $150 regardless of age. If we compare Platinum and Silver members, the rate of increase in purchase amount by age is higher for Platinum members than Silver members.

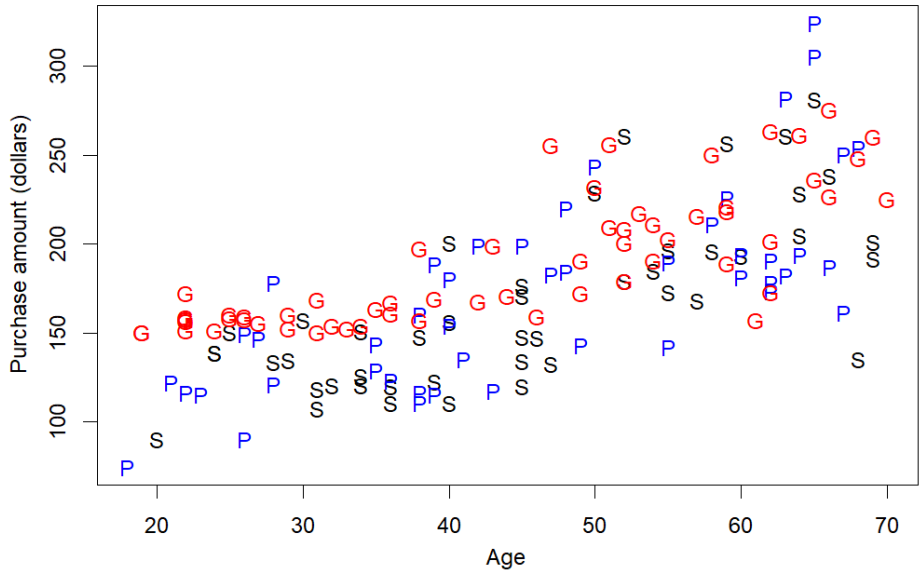


Figure 4: Scatter plot of purchase amount against age by membership level (G=Gold, P=Platinum, S=Silver).

1(c) Fitting the multiple linear regression model with both explanatory variables Age and Membership Level and an interaction term (Model 1), summary output includes

```
lm(formula = PurchaseAmount ~ Age * Level)

                    Estimate   Std. Error  t value  Pr(>|t|)
(Intercept)        106.4386    12.2925    8.659    8.77e-15 ***
Age                  1.8951     0.2640    7.179    3.45e-11 ***
Level1Platinum     -53.8151    19.4549   -2.766    0.00642 **
Level1Silver       -51.1505    20.1335   -2.541    0.01213 *
Age:Level1Platinum   0.7150     0.4072    1.756    0.08122
Age:Level1Silver     0.5398     0.4273    1.263    0.20849

Residual standard error: 31.2 on 144 degrees of freedom
Multiple R-squared:  0.5702,   Adjusted R-squared:  0.5553
F-statistic: 38.21 on 5 and 144 DF,  p-value: < 2.2e-16
```

The *F*–test indicates very strong evidence ($p < 2.2\times10^{-16}$) that this model fits better than the null model. That is, Age and Membership level together provide useful information about Purchase amount. However, the coefficient of determination R^2 is only 57.02%, indicating that some variation in the data remains unexplained. Looking at individual coefficients in the model, only Age and Membership level coefficients are significant (p=3.44×10^{-11} and p=0.0064, respectively), while those associated with the interaction terms are not statistical significance (p =0.0812 and p=0.2085) at the 5% significance level. We see that Age is highly significant ($p < 2.2\times10^{-16}$). With Age included, Membership level is also highly significant (p=2.97×10^{-5}); but with the two main effects included, the interaction term is not significant (p=0.181). We can simplify the model by removing this term.

Next, perform an Analysis of Variance for Model 1:

```
             Df  Sum Sq   Mean Sq  F value   Pr(>F)
Age           1  160806   160806   165.166   < 2e-16 ***
Level         2   21838    10919    11.215    2.97e-05 ***
Age:Level     2    3366     1683     1.728    0.181
Residuals   144  140199    974
```

We see that Age is highly significant ($p < 2.2\times 10^{-16}$). With Age included, Membership level is also highly significant ($p = 2.97\times10^{-5}$); but with the two main effects included, the interaction term is not significant ($p = 0.181$). We can simplify the model by removing the interaction term.

Now fit an additive Model 2 (with both main effects, but no interaction term):

```
lm(formula = PurchaseAmount ~ Age + Level)

               Estimate   Std. Error  t value  Pr(>|t|)
(Intercept)     90.4585   8.6427      10.466   < 2e-16 ***
Age              2.2587   0.1733      13.031   < 2e-16 ***
LevelPlatinum  -21.5582   6.1816      -3.487   0.000645 ***
LevelSilver    -27.1716   6.2114      -4.374   2.3e-05 ***

Residual standard error: 31.36 on 146 degrees of freedom
Multiple R-squared:  0.5599,   Adjusted R-squared:  0.5509
F-statistic: 61.91 on 3 and 146 DF,  p-value: < 2.2e-16
```

Compared to the full model, R^2 has only decreased slightly (1.03%) to 55.99%. The *F*–test remains highly significant ($p < 2.2\times 10^{-16}$). This confirms that this simpler model (Model 2) should be preferred, and we can drop the interaction term since all individual coefficients are now significant (p-values are all less than 0.000645).

We further investigate the Analysis of Variance table of Model 2:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Age	1	160806	160806	163.5	< 2e-16 ***
Level	2	21838	10919	11.1	3.24e-05 ***
Residuals	146	143564	983		

We observe that Age ($p < 2\times 10^{-16}$) and Membership level ($p = 3.24\times 10^{-5}$) are both highly significant, indicating their importance as predictors in explaining the variability of the purchase amount in Model 2.

Next, upon examining the residuals of Model 2 (see Figure 5), the plot of residuals versus fitted values shows no particular pattern, though there appears to be one potential outlier at the observation number 134. The normal QQ plot suggests a slightly right-skewed distribution of residuals, with a subtle upward curve above the straight line at the upper tail of the plot. However, this deviation can be considered minor, with no significant deviations from the normality assumption. The scale-location plot suggests some heteroscedasticity, with variance increasing as fitted values increase. The Cook's distance plot values show no overly influential points. In particular, observation 134 does not seem overly influential, as its Cook's distance value is small (0.1038), well below the threshold of 1. If we re-fit the model with observation 134 excluded, we find that the estimates of regression coefficients are not much changed, confirming that observation 134 is not influential, so there's no need to exclude it.

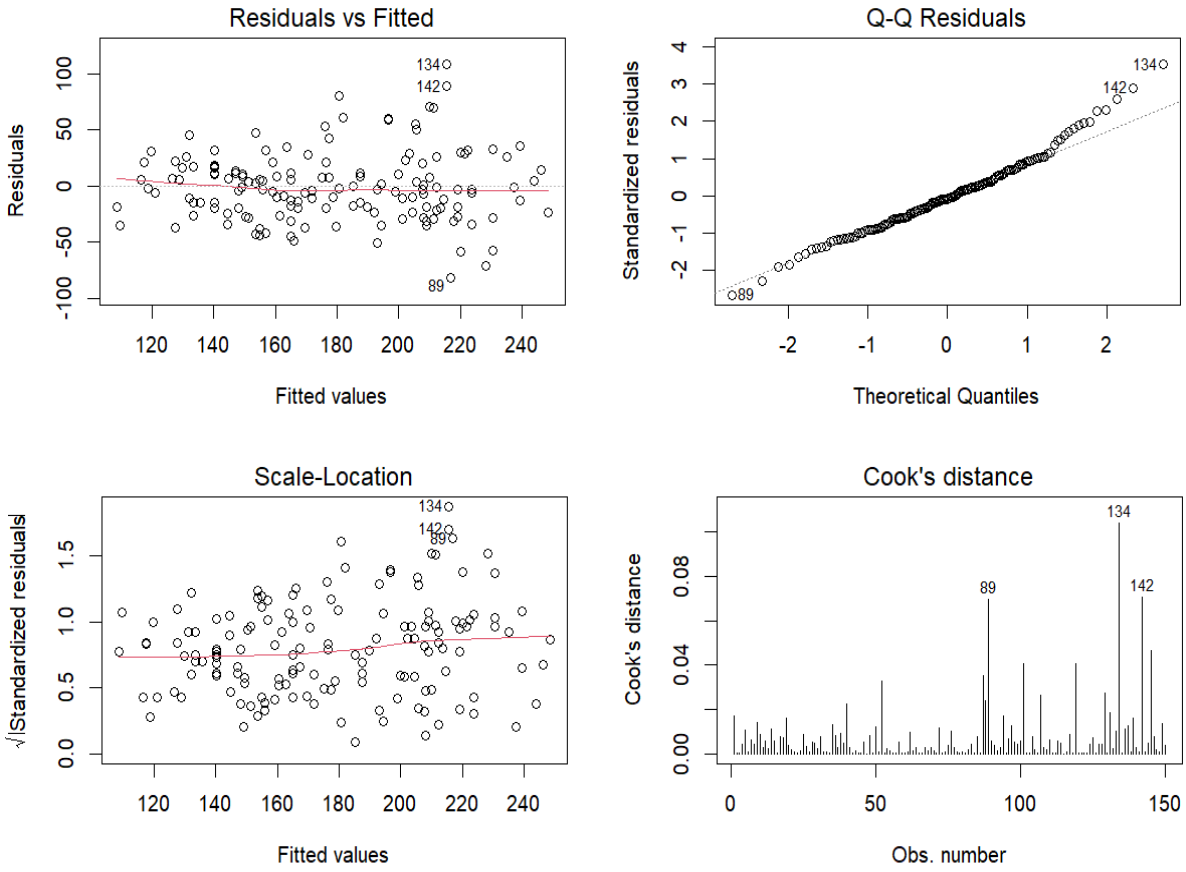


Figure 5: Residuals plots of Model 2.

Our preferred fitted model (Model 2) is

$$\text{PurchaseAmount}_i = 90.4585 + 2.2587(\text{Age}_i) - 21.5582 \times I\{\text{Leve1} = \text{Platinum}\}$$
$$- 27.1716 \times I\{\text{Leve1} = \text{Silver}\}$$

where $I(\,)$ is the indicator function of Membership Level with Gold as the base level.

Interpretation: The fitted Model 2 suggests that purchase amount increases by 2.2587 dollars for every one-year increase in age, holding membership level constant. Also, the negative coefficient values in the Platinum and Silver membership levels (–21.5582 and –27.1716, respectively) are associated with lower purchase amount compared to the Gold level. This is not surprising as those with Gold membership tend to spend more than those with Platinum and Silver (see Figure 6).

Conclusion: The preferred Model 2 provides valuable insights into the purchase amount since both the *F*–statistic and all coefficients *t*–tests are highly significant. The fitted model shows that the purchase amount increases as age increase. Individuals with Gold membership tend to purchase more than those with Platinum and Silver memberships. The $R^2 = 0.5599$ indicates that the model explains approximately 56% of the variability in the purchase amount using the explanatory variables age and membership level (without an interaction term). The plot of residuals versus fitted values is randomly scattered shows no issues with the assumptions of independence and mean 0. However, the assumptions of normality and homoscedasticity seem to be imperfectly met.

Discussion on the adequacy of model: While the fitted Model 2 provides some useful information about purchase amount, it also exhibits some obvious deficiencies. The R^2 value, approximately 0.5599, suggests that the model explains only around 56% of the variation in purchase amount (see Figure 6). This result is not surprisingly, considering we have only used two explanatory variables: age and membership level. Purchase amount could depend on other factors such as the purchased item, social grade, gender, etc. Additionally, the residuals Normal QQ-plot and the Scale-Location plots in Figure 5 indicate slight deviations from the normality and homoscedasticity assumptions. Moreover, the scatter plot in Figure 4 shows a slightly positive quadratic pattern than a linear one between purchase amount and age. Exploring a model with a quadratic term for age could potentially account for its nonlinearity and improve the model's performance.

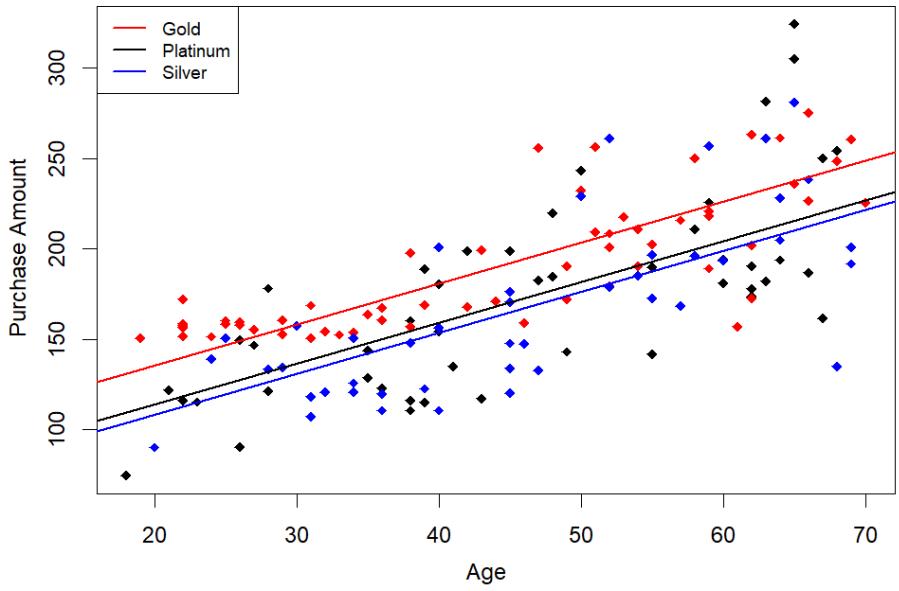


Figure 6: Scatter plot of purchase amount plotted against age, with Model 2 fitted lines representing membership levels.

1(d) (i) For a 30-year-old customer with Platinum, the estimated purchase amount is

$$\text{PurchaseAmount} = 90.4585 + 2.2587(30) - 21.5582 = \$\,136.661$$

(ii) For a 60-year-old customer with Gold membership, the estimated purchase amount is

$$\text{PurchaseAmount} = 90.4585 + 2.2587(60) = \$\,225.98$$

Suggestion: Based on the insights derived from the fitted values for 1(d)(i) and 1(d)(ii) and the preferred fitted equation discussed in 1(c) (see Figure 7), the model indicates that older individuals with Gold memberships tend to make higher purchases compared to younger individuals with Platinum memberships. Consequently, the manager could implement targeted marketing campaigns aimed at older age groups, promoting products or services tailored to their preferences and needs. Given that Gold members exhibit higher spending tendencies than Platinum members, the manager may consider introducing special promotions or loyalty programs to incentivize increased spending among Gold members. Additionally, the retail manager may also consider introducing targeted promotions, rewards, and special offers tailored to the preferences of Platinum members to encourage increased spending among them.

2. (a) The sample size is n =200. The numerical summary of age:

```
Min.    1st Qu.  Median   Mean    3rd Qu.  Max.
19.00   36.00    54.00    53.16   70.00    85.00
```

Sample standard deviation of age is 19.56.

As seen in Figure 7, the distribution of AGE closely approximates a normal distribution, with a mean of 53 and a standard deviation of 19. Both the median and mean values are close, suggesting a distribution resembling normality. The dataset ranges from a minimum age of 19 to a maximum age of 85 among the 200 patients.

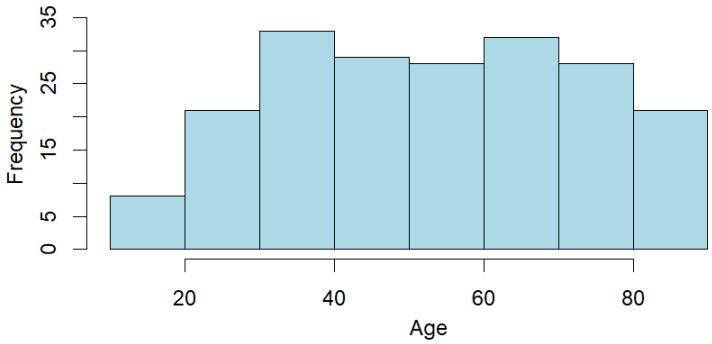


Figure 7: Histogram of age of patients.

For categorical variable STA, 185 patients were discharged from the hospital having 'lived', while 15 patients died (refer to Figure 8(a)). Regarding the medical conditions (MC), the observed frequencies are shown in Table 2: 40 patients were admitted to the hospital with arthritis, 30 with asthma, 31 with cancer, 33 with diabetes, 34 with hypertension, and 30 with obesity. The distribution of medical conditions (MC) across all six categories is relatively even, except for a slightly higher prevalence of arthritis (see Figure 8(b)).

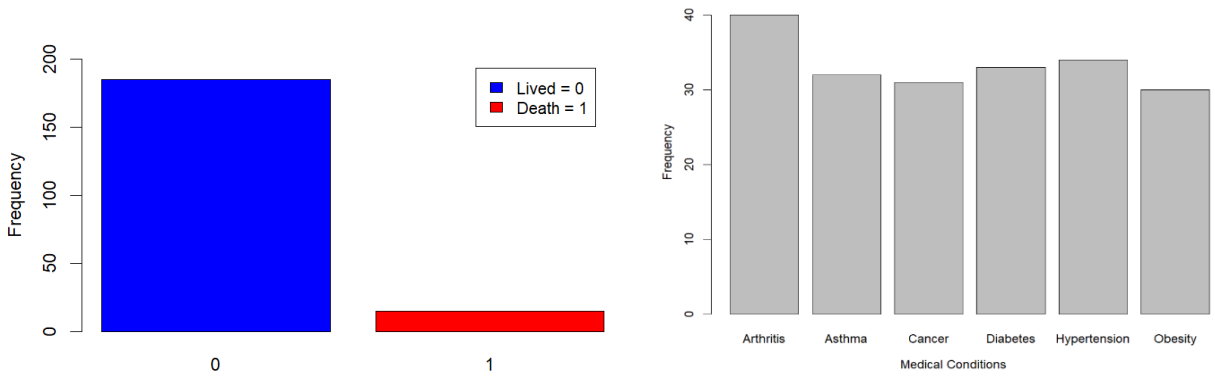


Figure 8: Bar plots of (a) status of hospital discharge (STA); (b) medical conditions (MC) for a sample of 200 adult patients.

Medical Conditions	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity
Frequency	40	32	31	33	34	30

Table 2: Frequency table of medical conditions (MC) of a sample of 200 adult patients.

2. (b) In Figure 9, the boxplot of age by vital status at hospital discharge shows the distribution of age for 'Lived=0', which approximates a normal distribution. However, for 'Death=1', the age distribution is noticeably skewed to the left, with an outlier on the left tail. This indicates that the 'Death=1' patient group tends to have an older age compared to the 'Lived=0' patient group.

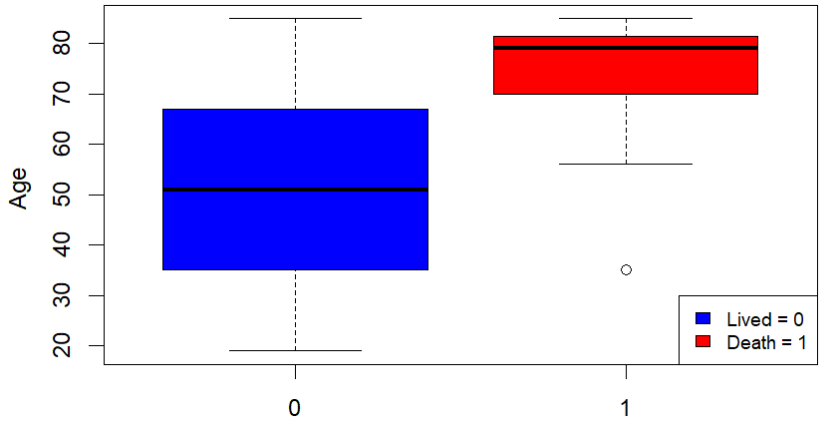


Figure 9: Boxplot of age (AGE) by vital status at hospital discharge (STA) for a sample of 200 patients.

2. (c) The table below displays the contingency table of vital status at hospital discharge (STA) and medical conditions (MC). As observed, some cells contain very small numbers, with some having zero counts. Due to limitations in conducting a chi-square test of independence arising from these low counts, Fisher's exact test appears to be a suitable alternative.

	Medical conditions (MC)					
STA	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity
'Lived = 0'	0	40	27	30	29	30
'Death = 1'	1	0	4	3	5	0

We use the Fisher test of independence to test:

H_0: STA and MC are independent
H_1: STA and MC are not independent

```
Fisher's Exact Test for Count Data

data:  contingency_table
p-value = 0.03126
alternative hypothesis: two.sided
```

Based on the *R* output above, the associated *p*-value for the Fisher's exact test is 0.03126, which is less than α=0.05. This suggests that the vital status at hospital discharge (STA) depends on the medical conditions (MC) at the 5% level of significance. However, it is worth noting that this relationship does not hold at the 1% significance level.

2. (d) A logistic regression would be appropriate here because the response variable STA is binary data (indicating whether the patient 'Lived=0' or 'Death=1'). The logistic regression model with STA as the response variable and AGE and MC as explanatory variables can be written as:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1(\text{AGE}_i) + \beta_2(\text{MC}_i)$$

where

- π_i represents the probability of the *i*-th patient being discharged as deceased;
- $\text{logit}(\pi_i)$ is the log odds of the *i*-th patient being discharged as deceased;
- β_0 is the intercept term of the model;
- β_1 is the coefficient for the AGE variable;
- β_2 is the coefficient vector for the MC variable, which is categorical and would be encoded into several dummy variables in the regression model;
- AGE_i is the age of the *i*-th patient;
- MC_i is the medical conditions of the *i*-th patient, with one category chosen as the reference group.

A logistic regression is specifically designed for binary or dichotomous outcome data, modelling the probability of an event occurring. Typically, the default category, coded as 0, represents the absence of the event of interest. However, the model predicts the probability of the event occurring, coded as 1, rather than focusing on the default category. In our case, the default category represents survival (STA=0).

In contrast, linear regression is unsuitable for binary outcome data as it assumes that the response variable is continuous and normally distributed. Linear regression cannot directly model binary outcomes and is not appropriate for estimating probabilities or predicting binary outcomes. Additionally, linear regression may produce predictions outside the [0, 1] range, which are not meaningful probabilities.

Another reason logistic regression is suitable because it utilizes the logit link function, ensuring that predicted values are valid probabilities lying between 0 and 1. If the predicted probability falls below a threshold probability value (typically 0.5, but subject to variation for imbalanced data or specific contexts), the observation is classified into the negative class (coded as 0); otherwise, it is classified into the positive class (coded as 1).

2. (e) Fitting a generalized linear model with a binomial response (STA) and the default link function of logistic regression, along with the two explanatory variables AGE and MC without interaction terms (Model A), yields the following output:

```
                Estimate  Std. Error  z value  Pr(>|z|)
(Intercept)    -26.04525  2483.16649  -0.010   0.99163
AGE              0.08921     0.02505   3.561   0.00037 ***
MCAsthma        17.84154  2483.16589   0.007   0.99427
```

```
MCCancer          19.05415  2483.16589   0.008   0.99388
MCDiabetes        17.72990  2483.16589   0.007   0.99430
MCHypertension    18.42552  2483.16587   0.007   0.99408
MCObesity         -0.13660  3823.76549   0.000   0.99997

Null deviance: 106.554  on 199  degrees of freedom
Residual deviance:  70.189  on 193  degrees of freedom
AIC: 84.189
```

The individual hypothesis test indicates that the explanatory variable AGE is highly significant (*p*-value = 0.00037). However, for the variable MC, none of the coefficients for the levels Asthma, Cancer, Diabetes, Hypertension, and Obesity are statistically significant when compared to the base level Arthritis, as the *p*-values exceed 0.99 for all these *t*-tests. These findings suggest that the categorical variable MC does not significantly contribute to explaining the variation in the response variable STA in Model A. Further investigation of the categorical variable MC is required. Additionally, the intercept also shows insignificance (*p*-value = 0.99163).

To further investigate the importance of the categorical variable MC, an Analysis of Deviance for Model A is performed, giving the following results:

```
     Df  Deviance  Resid.Df  Resid.Dev  Pr(>Chi)
NULL                  199      106.554
AGE   1    21.193     198       85.361    4.152e-06 ***
MC    5    15.172     193       70.189    0.009654 **
```

It shows that AGE is highly significant (*p*-value = 4.152×10^{-6}); in the presence of AGE, the inclusion of MC is also significant (*p*-value = 0.00965). The results suggest that both explanatory variables AGE and MC should be retained in the model.

Next, fitting models with each of the explanatory variables, separately, as well as the null model, we obtained the following residuals deviance and AIC values.

Terms present	Deviance	df	AIC
AGE + MC	70.189	193	84.189
AGE	85.362	198	89.361
MC	92.255	194	104.25
Null	106.55	199	108.55

Analysis of deviance:

Comparing AGE + MC model with AGE only model gives a chance in deviance of 85.362 – 70.189 = 15.172 on df = 198 – 193 = 5, with an associated *p*-value of $P(\chi_5^2 > 70.189) = 0.009654$. Thus, removing MC from the model does significantly reduce goodness-of-fit.

Comparing AGE + MC model with MC only model gives a chance in deviance of 92.255 – 70.189 = 22.066 on df = 194 – 193 = 1, with an associated *p*-value of $P(\chi_1^2 > 22.066) = 2.635\times10^{-6}$. Thus, removing AGE from the model results in a highly significant reduction in goodness-of-fit.

Comparing AGE + MC model (Model A) with the Null model, change in deviance is 106.55 – 70.189 = 36.365 on df = 199 – 193 = 6, with an associated *p*-value of $P(\chi_1^2 > 22.066) = 2.34\times10^{-6}$. So, very strong evidence that Model 1 with both variables AGE and MC fits better than the null model.

Comparing our preferred Model A with the maximal model, deviance is 70.189 on 193 degrees of freedom, so goodness-of-fit test gives p-value of $P(\chi^2_{193} > 70.189) = 1$ (to machine precision). Fit of Model A is not significantly worse than the maximal model.

Looking at AIC values,

$$\text{AIC}_{\text{AGE}} - \text{AIC}_{\text{AGE_MC}} = 89.361 - 84.189 = 5.172 > 2$$
$$\text{AIC}_{\text{MC}} - \text{AIC}_{\text{AGE_MC}} = 104.25 - 84.189 = 20.066 > 2$$
$$\text{AIC}_{\text{Null}} - \text{AIC}_{\text{AGE_MC}} = 108.55 - 84.189 = 24.365 > 2$$

we observe that models with one explanatory variable and the null model have AIC values much larger than Model A (the difference is larger than the rule-of-thumb of 2, representing a 'substantial' improvement in fit). Additionally, Model A has the lowest AIC value (84.189), indicating the best fit to data among all the four models investigated.

The preferred fitted model (Model A) is

$$\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = -26.05 + 0.089(\text{AGE}_i) + 17.84I(\text{MCAsthma}_i) + 19.05I(\text{MCCancer}_i) + 17.73I(\text{MCDiabetes}_i) + 18.43I(\text{MCHypertension}_i) - 0.14I(\text{MCObesity}_i)$$

where $\hat{\pi}_i$ is the estimated probability of the i-th patient being discharged as deceased, and $I(\,)$ is the indicator function of MC with Arthritis as the base level.

The positive coefficient for AGE (0.089) indicates that as the age of the patient increases, there is a corresponding increase in the probability of being discharged as deceased (STA=1).

Plotting deviance residuals against fitted values (Figure 10) shows two distinct groupings, with deviance residuals decreasing as fitted values increase within each group. However, this pattern is characteristic of the binary response variable STA and does not necessarily indicate a problem with the fit.

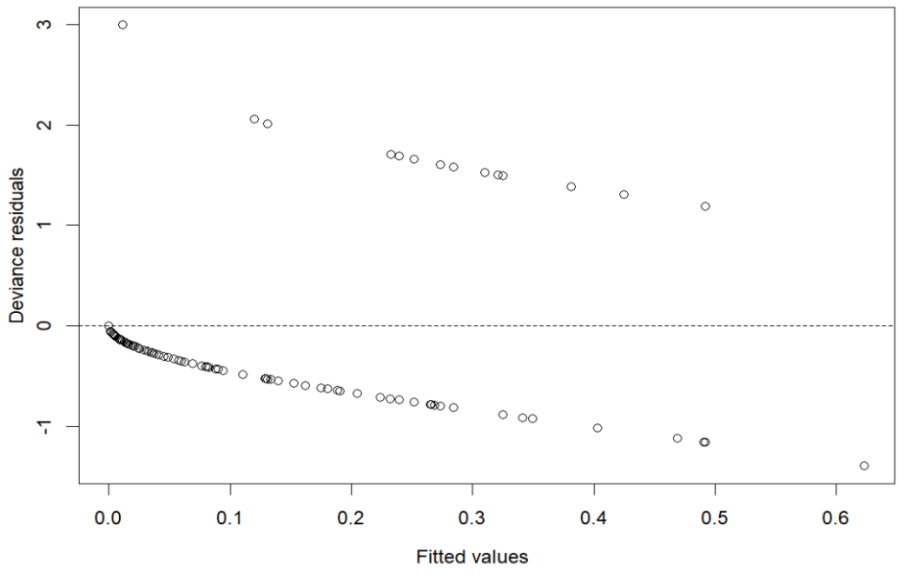


Figure 10: Deviance residuals against fitted values for Model A.

Conclusion: The analysis of deviance and AIC values suggest that both variables AGE and MC have a significant effect on the vital status at hospital discharge (STA); therefore, they should be retained in the model. The deviance residuals plot shows no significant model inadequacy. For preferred fitted Model A, the positive coefficient for AGE (0.089) indicates that the

probability of death (STA=1) increases with age. This relationship is also evident from the boxplots in Figure 9. Furthermore, the scatter plot in Figure 11 shows a linear relationship between the logit-transformed probability, $\text{logit}(\hat{\pi}_i)$, and AGE, confirming the suitability of Model A. The plot displays two clearly discernible sets of linear lines, reflecting the influence of the categorical variable levels of MC. Overall, the fitted Model A appears satisfactory.

Discussion: Collecting more data and including additional relevant variables might enhance the predictive power of the logistic regression model. Variables such as gender, types of medication, blood test results, socioeconomic factors, and severity of medical conditions could offer valuable insights into the outcome. Additionally, further exploring interactions between these variables may improve the fitted model.

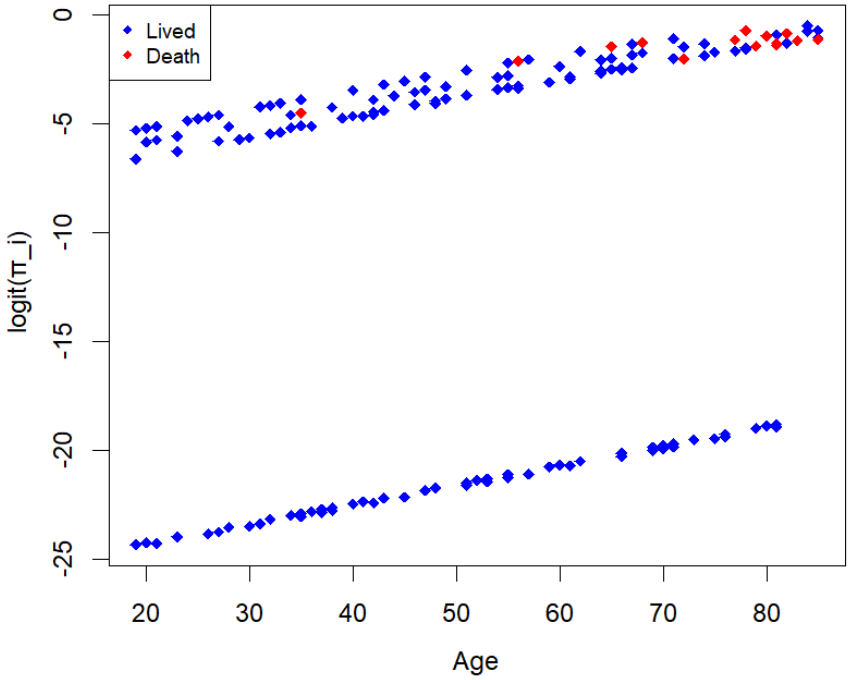


Figure 11: Scatter plot illustrates the linear relationship between the logit-transformed probability, $\text{logit}(\hat{\pi}_i)$, and AGE for Model A.

Rubric Exposition/Presentation

Overall exposition/presentation: 10 marks. Follow the rubric in Appendix B. Note that this rubric is written with the intention that 10 marks will only be awarded to an exceptionally good piece of work; expect most students to get 8 to 5 marks; expect that fewer students will get 0 to 4 marks.